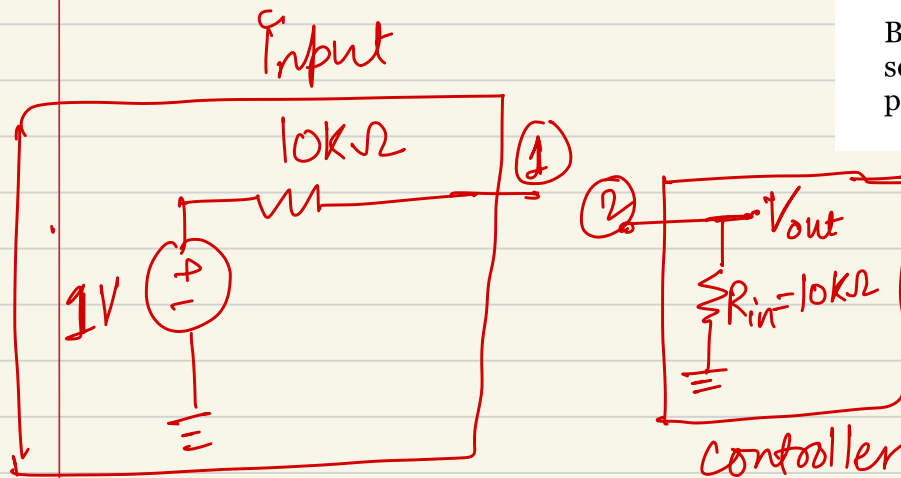


Tutorial 8 Solutions

Q1. You have a carbon-fibre strain gauge on a bridge that outputs a 1 V signal and has an internal resistance of 10 k Ω . You need to feed this signal into a digital controller that also has an input resistance of 10 k Ω .

A. If you connect the sensor directly to the controller, calculate the actual voltage the controller will read. What is the percentage of data lost?

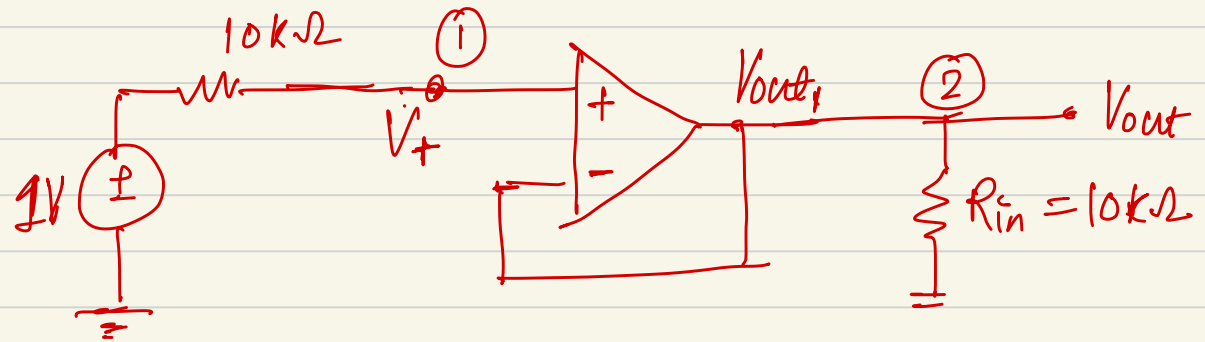
B. You insert an ideal non-inverting Op-Amp buffer (Voltage Follower) between the sensor and the controller. Calculate the new voltage read by the controller, and prove mathematically how the buffer fixes the problem.



A) If terminals ① & ② are directly connected.

$$V_{out} = \frac{10k}{10k + 10k} \times 1V = 0.5V \quad (\text{i.e. } 50\% \text{ reduced compared to input of } 1V)$$

(B)



Assuming current going inside the non-inverting terminal $\sim 0A$

$$\Rightarrow \text{voltage at ①: } V_+ = 1 - 10k\Omega \times 0 = 1V$$

Opamp voltage follower translates V_+ to $V_{out,1} = V_+$

$$\Rightarrow V_{out} = V_{out,1} = 1V \quad (\text{Assuming zero internal output resistance of the OpAmp})$$

Q2. A drone needs to combine data from three sensors: a gyroscope (v_1), an optical flow sensor (v_2), and an accelerometer (v_3). The drone's microcontroller has a limited number of physical input pins. To save connection pins and make the computer's job simpler, you decide to mix these three analog signals into one single signal *before* plugging it into the computer. Design a circuit to physically calculate this exact equation: $V_{\text{control}} = -(2v_1 - 4v_2 + 0.5v_3)$.

$$V_{\text{control}} = -2V_1 + 4V_2 - 0.5V_3$$

We know that an inverting summing amplifier can provide

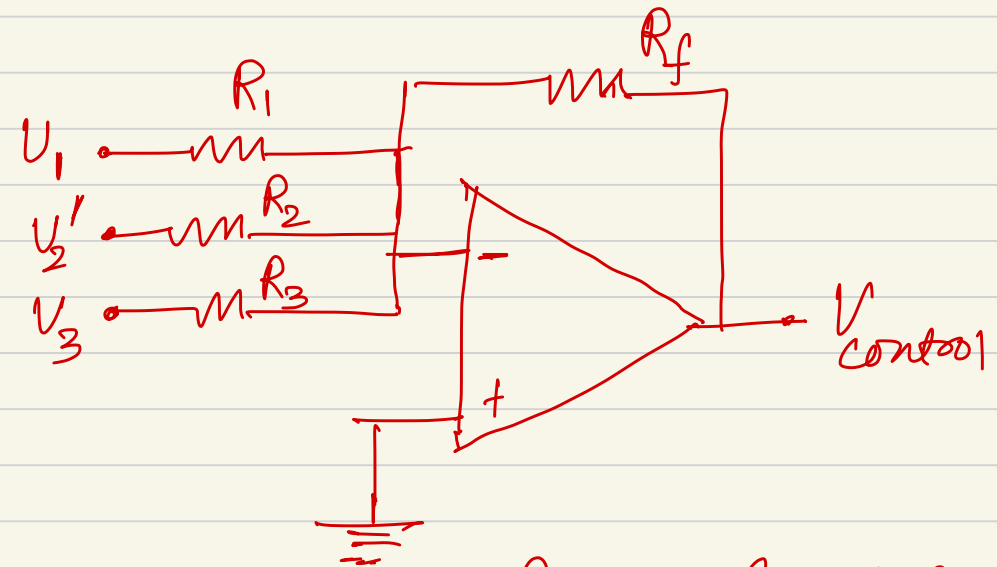
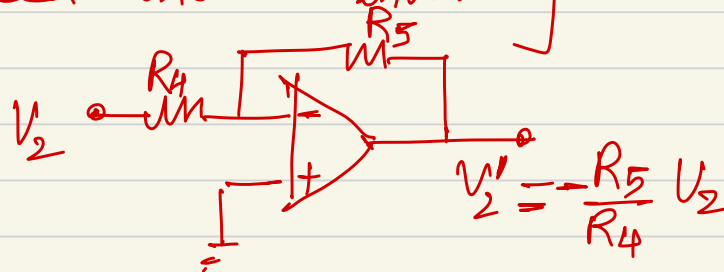
$$V_o = -(A V_A + B V_B + C V_C)$$

but as V_2 term is positive, we will require pre-processing of that signal.

$$\Rightarrow +\frac{R_f}{R_1} = +2 \quad \& \quad +\frac{R_f}{R_3} = +0.5$$

$$\Rightarrow R_1 = \frac{R_f}{2} \quad \& \quad R_3 = 2R_f$$

For V_2' , we need another inverting amplifier.

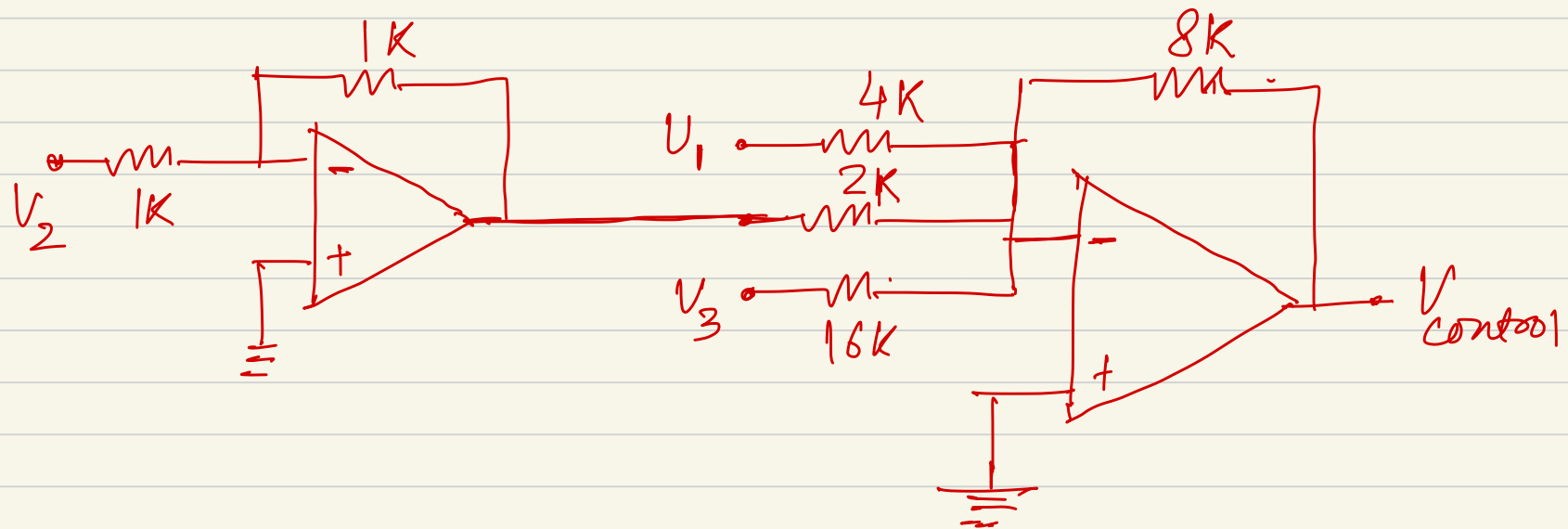


$$\Rightarrow V_{\text{control}} = -\frac{R_f}{R_1} V_1 - \frac{R_f}{R_2} V_2' - \frac{R_f}{R_3} V_3$$

$$= -\frac{R_f}{R_1} V_1 + \frac{R_f R_5}{R_2 R_4} V_2 - \frac{R_f}{R_3} V_3$$

$$\Rightarrow + \frac{R_f}{R_2} \times \frac{R_5}{R_4} = 4 \quad \Rightarrow \frac{R_2 R_4}{R_5} = \frac{R_f}{4}$$

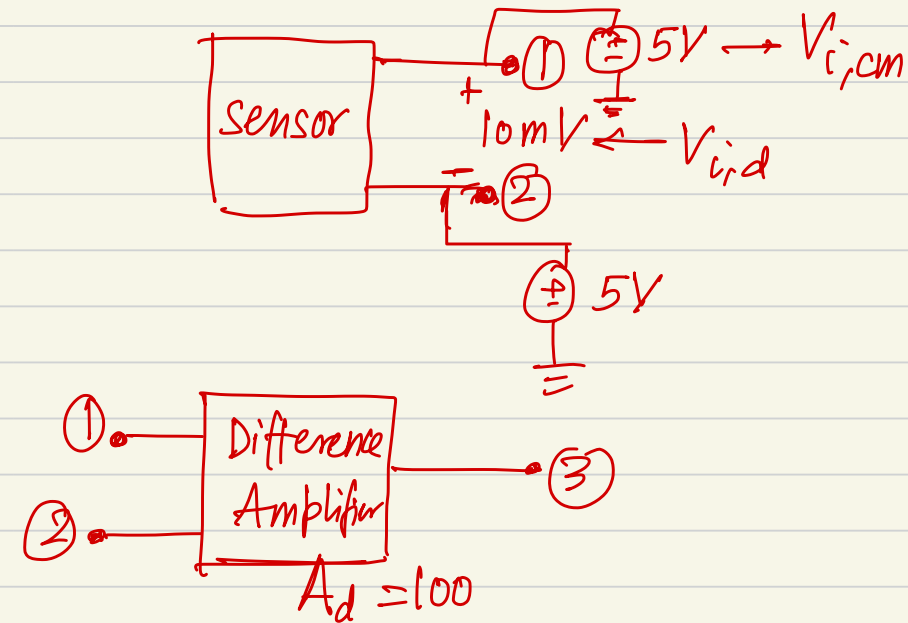
Choosing $R_f = 8k\Omega$, $R_1 = 4k\Omega$, $R_3 = 16k\Omega$
 if $R_4 = R_5 = 1k\Omega$, $R_2 = 2k\Omega$



Q3. You are designing the closed-loop temperature control system for a plasma etching chamber in the TATA's semiconductor foundry. The silicon wafer's temperature is monitored by a precision differential sensor that outputs a **tiny 10 mV signal (V_d)**. However, the high-power RF generator used to strike the plasma is located right next to the sensor array. This generator induces a massive **5 V common-mode noise signal (V_{cm})** equally onto both of your delicate sensor wires. Your receiver circuit uses a difference amplifier with a built-in differential gain of $A_d = 100$.

A. If you use a basic Op-Amp with a Common-Mode Rejection Ratio (CMRR) of **40 dB**, calculate the exact common-mode gain (A_{cm}) and the total voltage seen at the output (Signal + Noise). Explain why this will ruin the wafer batch.

B. You upgrade the board to a precision instrumentation amplifier with a CMRR of **100 dB**. Calculate the new total voltage at the output and prove mathematically why the controller can now do its job.



A. If diff. amp, $CMRR = 40 \text{ dB} = 10^2$

$$\Rightarrow CMRR = \frac{|A_d|}{|A_{cm}|} \Rightarrow A_{cm} = \frac{100}{100} = 1$$

$$\Rightarrow A_{cm} = \frac{V_{o,cm}}{V_{i,cm}} \Rightarrow V_{o,cm} = A_{cm} V_{i,cm}$$

$$V_{i,cm} = 5 \text{ V}$$

$$\Rightarrow V_{o,cm} = 5 \text{ V}$$

$$A_d = \frac{V_{o,d}}{V_{i,d}} = 100$$

$$V_{i,d} = 10 \text{ mV}$$

$$\Rightarrow V_{o,d} = 10 \text{ mV} \times 100 = 1 \text{ V}$$

$$\Rightarrow \text{Total voltage at (3), } V_o = V_{o,d} + V_{o,cm} = 6 \text{ V}$$

→ As the output is dominated by amplified noise output, it corrupts the control.

B. When upgraded to CMRR of 100 dB,

$$\Rightarrow \text{CMRR} = \frac{A_d}{A_{cm}} = 10^5$$

$$\Rightarrow A_{cm} = \frac{100}{10^5} = 10^{-3} = \frac{V_{o,cm}}{V_{i,cm}}$$

$$\Rightarrow \boxed{V_{o,cm} = 5 \text{ mV}}$$

$$\begin{aligned} \Rightarrow V_o &= V_{od} + V_{o,cm} \\ &= 1 + 0.005 \\ &= 1.005 \text{ V} \end{aligned}$$

$\Rightarrow$ close to the expected $V_o = 1 \text{ V}$

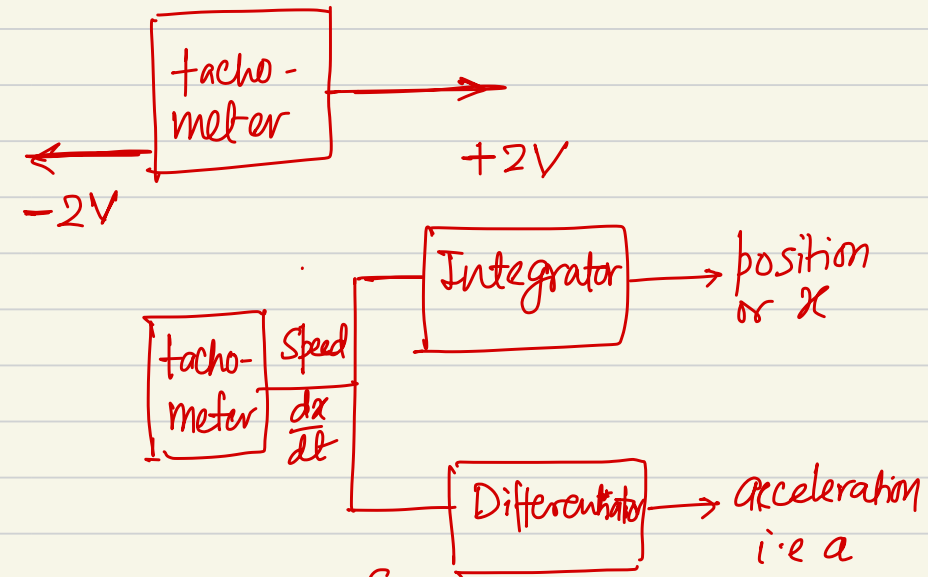
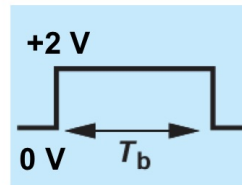
Q4. You are designing a budget robot vacuum. To save money, the front bumper switches were removed. The only sensor left is a tachometer (speed monitor) on the wheels that outputs $+2\text{V}$ when driving forward, and -2V when driving in reverse. You split this speed signal and feed it into two different Op-Amp circuits to help the robot navigate.

A. You feed the speed signal into an **Op-Amp Integrator**. What physical property of the robot (position, speed, or acceleration) does this output represent? Plot the output for the given input. Assume zero initial conditions across the capacitor.

B. You feed the exact same speed signal into an **Op-Amp Differentiator**. What physical property does this output represent? Plot the output for the given input. Assume zero initial conditions across the capacitor.

C. To design the resistor and capacitor values for these circuits, you are given following scenarios:

1. It takes the robot exactly 10 seconds to drive across the room. You want the **Integrator** to output exactly -10V after 10 seconds of driving forward at $+2\text{V}$. If you choose a standard capacitor of $C=10\text{ }\mu\text{F}$, calculate the required value of the resistor (R). Assume OpAMP DC Supplies of $\pm 12\text{ V}$.
2. When the robot smashes into the wall, it violently switches from driving forward ($+2\text{V}$) to driving backward (-2V). This sudden change takes just **0.1 seconds**. You want the **Differentiator** to output a sharp $+5\text{V}$ spike when this happens to act as a crash alarm for the computer. If you choose a capacitor of $C=1\text{ }\mu\text{F}$, calculate the required value of the resistor (R).



A.

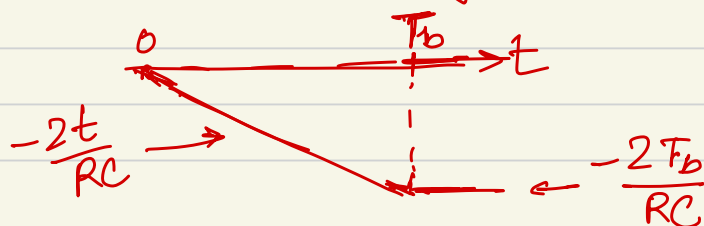
$$V_{out} = -\frac{1}{RC} \int V_{in} dt + V_{out}(0)$$

$$V_{out} = -\frac{1}{RC} \int_0^t +2V \cdot dt \quad (0 < t < T_b)$$

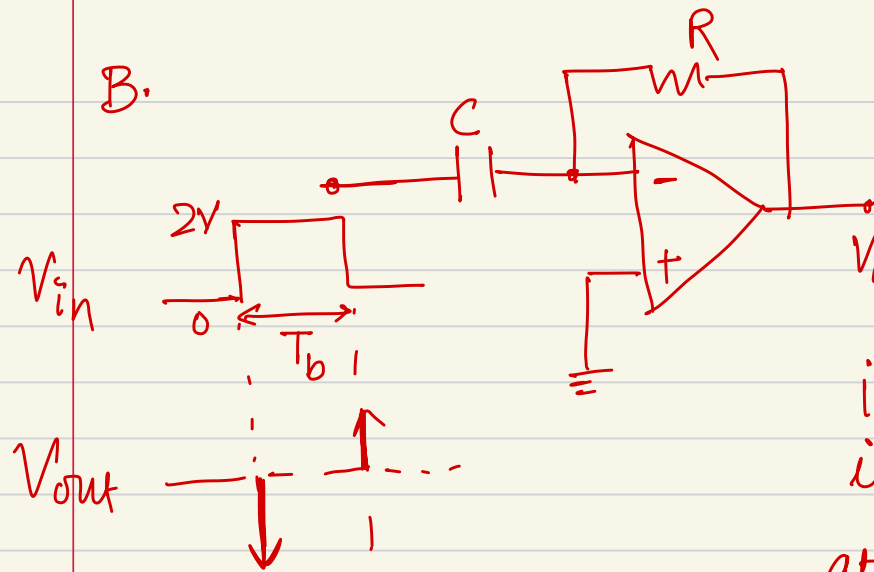
$$= -\frac{2t}{RC}$$

at $t = T_b$, $V_{out}(T_b) = -\frac{2T_b}{RC}$
 for $t \gg T_b$, $V_{out} = -2T_b/RC$

Output of Integrator:-



B.



$$V_{out} = -RC \frac{dV_{in}}{dt}$$

i.e. output will be finite where there is a change in the input.

at $t = 0$, input changes from 0 to 2V in $\Delta t \rightarrow 0$

$$\Rightarrow V_{out} = -RC \frac{(2-0)}{\Delta t} \rightarrow \text{impulse with negative magnitude}$$

at $t = T_b$, input changes from 2V to 0 in $\Delta t \rightarrow 0$

$$\Rightarrow V_{out} = -RC \frac{(0-2)}{\Delta t} = + \frac{2RC}{\Delta t \rightarrow 0}$$

$\rightarrow$ impulse with positive magnitude

C. 1. $V_{out} = -\frac{2t}{Rc}$, given $V_{out} = -10V$ at $t = 10\mu s$
 $C = 10\mu F$

$$\Rightarrow R = \frac{-2t}{V_{out} \cdot C} = + \frac{2 \times 10}{+10 \times 10 \times 10^{-6}} = \frac{2}{10} \times 10^6 = 200k\Omega$$

2. $V_{out} = -RC \frac{dV_{in}}{dt}$ $\xrightarrow[\text{small}]{\text{if } dt \text{ is}}$ $V_{out} = -RC \frac{\Delta V_{in}}{\Delta t}$

$\Delta t = 0.1\mu s$, $V_{in} : +2V \rightarrow -2V$

$\Rightarrow \Delta V_{in} = -4V$

expected $V_{out} = +5V$

given $C = 1\mu F$

$$\Rightarrow +5V = +R \times 10^{-6} \frac{(+4)}{0.1}$$

$$\Rightarrow R = \frac{0.5 \times 10^6}{4} \Omega$$

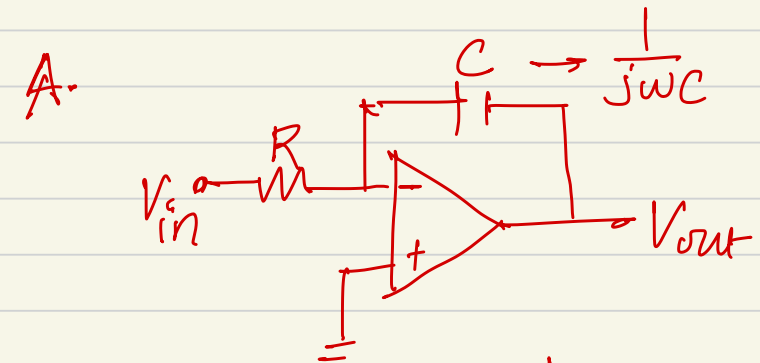
Q5. You are building an EEG monitor to read a patient's brainwaves. The target brainwave is a tiny, alternating signal at 10 Hz. However, you face the following two major signal problems:

- The metal sensors on the patient's skin produce a constant, **steady +0.1 V** (i.e. 0 Hz DC voltage).
- The long sensor cables act like antennas and pick up **high-frequency 60 Hz electrical noise** from the room's lighting.

- A. You first build a basic Integrator using an input resistor $R_{in} = 10 \text{ k}\Omega$ and a feedback capacitor $C = 100 \text{ nF}$. What is the theoretical gain of this circuit at 0 Hz? What will eventually happen to the Op-Amp's output if it tries to integrate that constant +0.1 V skin voltage? Assume OpAMP open loop gain to be 100 dB.
- B. To save the circuit, you add a feedback resistor $R_f = 100 \text{ k}\Omega$ in parallel with the capacitor. Calculate the newly stabilized output voltage caused by the +0.1 V skin sensor.
- C. Plot the **Gain with frequency** for the modified circuit. Calculate the **cutoff frequency (f_c)**. Does this cutoff frequency successfully allow the 10 Hz brainwave to pass through while blocking the 60 Hz high-frequency room noise?

given: -

V_{in} : 10 Hz signal
+ 0.1 V (dc offset)
+ 60 Hz noise



$$\Rightarrow V_{out} = -\frac{1}{j\omega RC} V_{in}$$

$$R = 10 \text{ k}\Omega, C = 100 \text{ nF}$$

$$\Rightarrow RC = 10^4 \times 100 \times 10^{-9} = 1 \text{ ms}$$

at 0 Hz,

$$A = \left| \frac{V_{out}}{V_{in}} \right| = \left| \frac{1}{\omega RC} \right| \rightarrow \infty$$

i.e. Capacitor behave like an open circuit
so the gain is limited by open loop
gain = 10^5 V/V .

$$\Rightarrow V_{out} = -0.1 \times 10^5 = -10^4 \text{ V}$$

However, V_{out} will be limited by DC
Supplies $\Rightarrow \boxed{V_{out} = -V_{cc}}$

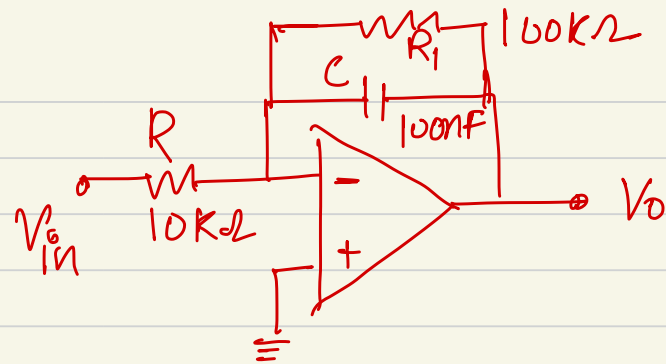
B.

$$V_o = -\frac{Z_2}{Z_1} V_{in}$$

$$= -\frac{R_1 \parallel \frac{1}{j\omega C}}{R} V_{in} = -\frac{R_1}{R} \frac{V_{in}}{1+j\omega CR_1}$$

for $V_{in} = +0.1V$ (DC)

$$V_o = -\frac{R_1}{R} V_{in} = -\frac{100k}{10k} (+0.1V) = -1V$$



$$|A| = \frac{R_1}{R} \frac{1}{\sqrt{1+(\omega/\omega_c)^2}}$$

where, $\omega_c = \frac{1}{R_1 C}$

C.

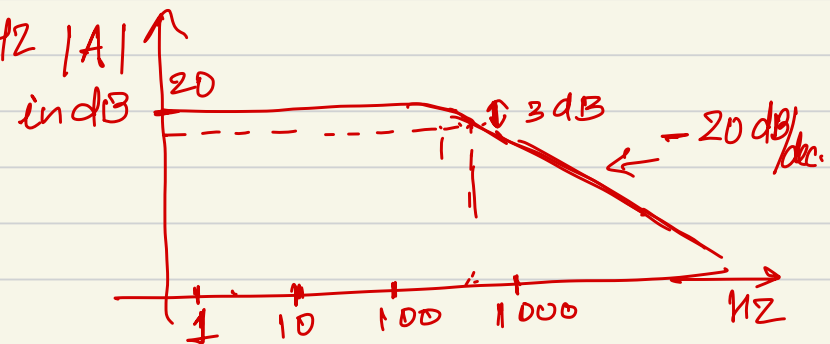
Cut-off freq. $\omega_c = \frac{1}{R_1 C} = \frac{1}{100 \times 10^3 \times 100 \times 10^{-9}} = \frac{1}{10^{-2}} = 100 \text{ rad/s}$

$$\Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{100}{2\pi} \approx 15.92 \text{ Hz}$$

$|A_1| = 10 \text{ V/V}$ at $\omega=0$ or 20 dB

$|A_2| = 17 \text{ dB}$ at $f = f_c$

for $f > f_c$, $|A|$ decreases by -20 dB/dec .



→ Yes, it will pass all the frequencies till ~ 16 Hz with a gain of 20 dB.